Zizhe Zhang

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EDUCATION

Master of Science in Engineering (M.S.E.) in Robotics Aug 2024 – Present
University of Pennsylvania, GRASP Lab *Philadelphia, PA*
Advisor: Prof. Nadia Figueroa

Visiting Student Jan 2023 – Mar 2023
University of California San Diego, Jacobs School of Engineering *San Diego, CA*

Bachelor of Engineering (B.E.) in Measuring Control Technology & Instruments Sep 2017 – May 2021
Southeast University, School of Instrument Science and Engineering *Nanjing, China*
Advisor: Prof. Yuan Yang

PUBLICATIONS

(* and † denote equal contribution)

VLMgineer: Vision Language Models as Robotic Toolsmiths
George Jiayuan Gao*, Tianyu Li*, Junyao Shi, **Zizhe Zhang**†, Yihan Li†, Nadia Figueroa, and Dinesh Jayaraman
Under Review at Conference on Neural Information Processing Systems (NuerIPS), 2025 [Website]
*Robotics: Science and Systems (RSS) Workshop on Robot Representations (**Spotlight Presentation**)*, 2025

Image-Based Visual Servoing for Enhanced Cooperation of Dual-Arm Manipulation
Zizhe Zhang, Yuan Yang, Wenqiang Zuo, Guangming Song, Aiguo Song, and Yang Shi
IEEE Robotics and Automation Letters (RA-L), 2025 [arXiv] [PDF] [Website]

RESEARCH AND INDUSTRY EXPERIENCE

Visiting Scholar, Duke University *Durham, NC*
Robot Dexterity Lab, advised by Prof. Xianyi Cheng Jun 2025 – Present
Working on developing an adaptive compliance motion representation for generalizable robotic manipulation and physical human–robot interaction (In Progress)

Graduate Research Assistant, University of Pennsylvania *Philadelphia, PA*
Figueroa Robotics Lab, advised by Prof. Nadia Figueroa Nov 2024 – Present
Working on machine learning applications in control theory and robot manipulation. Projects include:
- Ensuring feasibility of passivity-based torque control via learned kinematic constraint mappings (*In Progress*)
- Implementing vision-based learning with force feedback for contact-rich manipulation tasks (*In Progress*)
- Utilizing Vision Language Models (VLMs) for robot policy synthesis (*RSS Workshop 2025*)

Technical Intern, Schneider Electric *Shanghai, China*
Working in the Kylin project, including: Jun 2023 – Aug 2023
- IGBT thermal simulations
- Capacitor lifetime assessments
- EMC testing
- Circuit design

Undergraduate Research Assistant, Southeast University *Nanjing, China*
Robotic Perception and Control Lab, advised by Prof. Yuan Yang Dec 2023 – Aug 2024
Worked on developing visual-servoing-based robotic control architectures. Projects include:
- Designed an enhanced dual-arm collaborative control system based on image-based visual servoing (*RA-L 2025*)
- Designed a shared teleoperation system control based on visual servoing (*Bachelor Thesis*)

Automotive Safety Technology Lab, advised by Prof. Dong Wang Mar 2022 – Jun 2023
- Applying edge detection to classify Martian topography and identify soft-ground hazards

- Designed a wheeled-leg ground-detection mechanism for rover traversal and analyzing force signals to predict mobility

Advanced Navigation Technology Lab, advised by Prof. Xuanpeng Li — Jul 2023 – Sep 2024
- Utilizing the PYTS library for signal visualization and building a CNN to extract and classify features from ADS-B radio signals

COURSE PROJECTS

News Source Classification, University of Pennsylvania — *Philadelphia, PA*
Team of 3, Leader — Aug 2024 – Dec 2024
- Collected and preprocessed 415,343 headlines from Fox News and NBC using sitemap scraping and BeautifulSoup, ensuring a balanced, cleaned dataset
- Built a TF-IDF + Logistic Regression baseline and fine-tuned BERT and DeBERTa models, applying k-fold cross-validation to eliminate data leakage
- DeBERTa K-fold achieved 88.66 % test accuracy, significantly surpassing the baseline (67.61 %)

Autonomous Vehicle based on GPS & DoF Camera, University of California San Diego — *San Diego, CA*
Team of 3 — Jan 2023 – Mar 2023
- Utilized Python and VESC to control the robot, DoF camera to find and track lanes, centimetric GPS and PID method to record and follow paths
- Brought the robot to a complete stop by using PyVesc and DepthAI libraries to run stop sign detection on the camera
- Enabled the robot to respond correctly to speed limit signs by performing text detection on the camera

Design and Implementation of a Weather Query and Display Module, Southeast University. — *Nanjing, China*
Team of 3, Leader — Jan 2023 – Mar 2023
- Used ASM to design the function and organize each part of the MCU. Designed a Weather Query and Display Module which can choose and display weather information of several customized cities
- The whole project is based on a small device which can be used as smart home device

HONORS AND AWARDS

Outstanding Bachelor's Thesis, Southeast University — 2024
Merit Student, Southeast University — 2021

SERVICE

Reviewer
IEEE Transactions on Robotics (T-RO) — 2025

SKILLS

Programming C, C++, Python, MATLAB
Tools CMake, PyTorch, ROS, RLBench, ManiSkill, CoppeliaSim, PyBullet, Issac Gym/Sim/Lab
Robots UR3, UR3e, Franka Emika Panda, Franka Research 3